

Swapnil Sanjay Joiode

Data Engineer

United States | +1 617-961-9478 | swapniljoiode22@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

Professional Summary

Data Engineer with **3+ years** of experience designing and maintaining ETL pipelines, data warehouse solutions, and scalable data workflows using SQL, Python, Airflow, Snowflake, and Databricks. Experienced in writing complex SQL queries, scripting automated data processing workflows, and implementing data quality measures to maintain data integrity across **2 - 3M+** records monthly. Skilled in data modelling, pipeline designs, workflow orchestration, and cross-functional teamwork with analysts, engineers, and stakeholders across logistics, retail, and finance domains. Brings strong backend data and analytics engineering expertise to support delivering reliable, analytics-ready data platforms and scalable data flows.

Work Experience

Data and Analytics Engineer | [Kalven Group LLC](#) | United States

Dec 2025 - Present

- Designed and maintained scalable ETL pipelines and data flows using Airflow, Snowflake, Databricks, and AWS to process **2-3M** logistics records monthly, reducing reconciliation defects from **70** to fewer than **15** per cycle.
- Developed reusable dbt models, dimensional schemas, and data warehouse tables supporting **50+** KPI definitions across logistics and finance reporting.
- Optimised Spark and AWS Lambda ingestion workflows, reducing pipeline runtime from **90 to 55** minutes while improving operational stability and data reliability.
- Wrote complex SQL queries and Python scripting workflows to transform and aggregate large operational datasets across **200+** hubs and routes, enabling routing improvements that reduced late deliveries by **18%**.
- Implemented automated quality measures, validation checks, schema testing, and freshness monitoring using SQL, Python, and dbt, reducing manual QA effort from **6-7** hours to **2-3** hours per cycle while maintaining data integrity.
- Collaborated with analysts, engineers, and business stakeholders to deliver scalable analytics-ready data products for **40+** end users while communicating project progress and technical updates across teams.
- Documented pipeline designs, transformation workflows, KPI definitions, and ETL processes to support governance, maintainability, and future scalability.

Data Analyst and Power BI Developer | [Vision Group Retail Inc.](#) | United States

Aug 2024 - Nov 2025

- Redesigned Snowflake dimensional models, semantic-layer datasets, and data warehouse tables, reducing model size from **10 GB to 1 GB** and refresh time from **10 to 2** minutes across **500K+** records.
- Developed dbt transformation workflows, Python ETL pipelines, and curated data marts supporting merchandising and supply chain analytics requirements.
- Implemented SQL queries, Python scripting, and automated validation checks across **500K+** records per refresh cycle, reducing source discrepancies from thousands to single-digit mismatches and improving data integrity.
- Delivered governed reporting datasets and standardised **50+** business metrics while automating CI/CD workflows that eliminated **12+** manual reporting processes per quarter.
- Collaborated with cross-functional stakeholders to support reporting requirements, prioritizing multiple analytics tasks and maintaining reliable data delivery timelines.
- Documented semantic models, KPI definitions, ETL workflows, and reporting processes to support audit readiness, transparency, and long-term maintainability.

Data Analyst | [TekSolve IT Solutions Inc.](#) | United States

Aug 2023 - Dec 2023

- Designed automated SQL and Python validation frameworks across **50K+** monthly HR and payroll records, reducing report error rates from **0.58% to 0.18%**.
- Developed SQL queries, scripting workflows, and data quality checks that identified discrepancies **3-5** days earlier than manual reviews, improving on-time delivery of monthly reporting packs.
- Built star-schema data models and reporting solutions for HR and finance stakeholders, reducing dashboard rebuild requests by **40%**.
- Performed data validation, cleansing routines, and quality monitoring to resolve source mismatches, nulls, and duplicate records while maintaining data integrity and reporting reliability.
- Collaborated with HR and finance teams to gather reporting requirements, prioritize multiple tasks, and communicate data issues and resolutions effectively.
- Documented ETL logic, validation rules, reporting definitions, and data flows to support scalable and reproducible reporting processes.

Senior Business Analyst | [Quantum Phinance Consulting Pvt. Ltd.](#) | India

Jan 2020 - Dec 2021

- Developed automated reporting workflows and KPI-driven analytics across approximately **100** financial assets, improving decision-making efficiency by **30%**.
- Collaborated with finance and risk stakeholders to gather reporting requirements and deliver accurate MIS and risk reporting solutions.
- Streamlined reporting processes, workflow documentation, and data validation activities across **20+** BRDs, FRDs, and SOPs, reducing clarification cycles from **4-5** requests to fewer than **2**.
- Performed quality assurance checks and resolved reporting discrepancies, reducing rebuild efforts by **25%** while maintaining reporting accuracy and data reliability.

Education

M.S. Data Analytics (Information Management) | [Clark University](#) | United States

Jan 2022 - Dec 2023

- **Grade:** 3.4/4.0
- **Relevant Modules:** Database Management, Machine Learning, Statistics, Data Visualisation, Data Warehousing, Predictive Analytics

B.E. Electronics and Telecommunication Engineering | [Mumbai University](#) | India

Aug 2015 - Jul 2019

- **Grade:** 6.7/10.0
- **Relevant Modules:** Database Systems, Programming Fundamentals, Data Structures, Probability and Statistics, Software Engineering

Projects

Fashion Retail Medallion Pipeline | Individual Case Study

Jun 2025

- Built an end-to-end medallion pipeline (bronze, silver, gold) on a fully free-tier stack using Python, dbt, Snowflake, DuckDB, and Cloudflare R2, with a dual warehouse path enabling permanent, zero-cost reproducibility.
- Modeled silver and gold layers in dbt with tests, model contracts, and lineage docs, designing 6 conformed dimensions and 5 fact tables, orchestrated as an idempotent Airflow DAG via astronomer-cosmos.
- Produced 11 modeled tables across 5 business-ready data marts feeding a Next.js decision dashboard on Vercel, on a containerized, GitLab CI/CD-governed foundation guaranteeing identical runs at zero infrastructure cost.

Valorant Gameplay Data Engineering Pipeline | Individual Case Study.

Feb 2025

- Built an end-to-end three-layer data pipeline (Presentation to Logic to Data) using Python, Pandas, and REST APIs, integrating a custom Valorant API client to ingest and normalize game metadata into structured dimension tables.
- Engineered a stochastic match simulation engine generating synthetic timelines of 25 rounds per match with combat events and player metrics, exported to multi-table CSV with custom exception handling and logging.
- Automated generation of 730+ synthetic matches per year across 4 round- and player-level tables, producing an ML-ready dataset supporting 6 planned use cases including win-probability, anomaly detection, and rank forecasting.

Skills

- **Data Engineering:** ETL Pipelines, ELT Pipelines, Data Pipelines, Pipeline Designs, Workflow Orchestration, Apache Airflow, ETL Tools, Batch Ingestion, Data Integration, Data Processing, Data Flows, Pipeline Automation
- **Data Warehousing and Modelling:** Snowflake, Databricks, dbt, Data Warehouse, Data Warehouse Tables, Data Modelling, Dimensional Modelling, Semantic Layer, Star Schema, PySpark, Apache Spark
- **Programming and Querying:** Python, SQL, SQL Queries, Scripting, Programming Language, Snowflake SQL, Databricks SQL, Pandas, Scikit-learn, Statsmodels
- **Data Quality and Governance:** Data Validation, Data Quality Checks, Quality Measures, Schema Testing, Freshness Monitoring, Regression Testing, Automated Validation Frameworks, Data Integrity, SLA Monitoring, Data Governance, Audit Controls, RBAC, CDC
- **Soft Skills:** Stakeholder Management, Cross-functional Collaboration, Teamwork, Problem Solving, Communication, Communicating Technical Concepts, Requirements Gathering, Technical Documentation, Prioritizing Multiple Projects, Analytical Thinking, Time Management

Certifications

- [Data Engineer Associate](#) | DataCamp
- [ETL Testing](#) | Udemy
- [Data Warehousing](#) | Udemy

Mar 2025 - Mar 2027

Apr 2024

Feb 2024